Problem

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In this practice problem you derive a differential equation for yet another problem. This time, we do not specify the values of the different quantities, so the model and the results stay general. First, the problem.

Mixing problem

A small river, the $River_{in}$, feeds into a lake and at the other side of the lake, the water flows through into another river, the $River_{out}$, that flows on to the sea. The water of the $River_{in}$ contains a certain pollutant. A factory along the lake shore is planning to dump more of the same pollutant into the lake.

How much would the factory contribute to the pollution of the lake? Does it add only a few percent, or much more?

Quantifying time and amount of pollutant

The amount of pollution can be quantified in various ways: in kg, in mol, in a concentration of either.... Here we will simply choose kg. An appropriate unit for the time t is days. So we define:

t: time, with $[t] = \text{day}$.

$Z(t)$: the total mass of the pollution dissolved in the lake, with $[Z] = \text{kg}$.

We assume that the water in the lake mixes well, so the pollution of the factory is distributed quickly over the lake. With quickly, we do not actually mean in a nano-second, but say, in half an hour or so: much quicker than in a day, so we can assume that the water in the lake is *ideally* mixed, i.e. on the timescale of the mathematical model, instantaniously.

First, we define the quantities for the river feeding into the lake, for the lake itself, and for the river flowing out of the lake:

- Q_{in}: the volumetric flow rate of $River_{in}$ in $\frac{m^3}{day}$.
- C_{in}: the concentration of pollution of $River_{in}$ in $\frac{kg}{m^3}$.
- V_{lake}: the volume of the lake in m^3.
- Q_{out}: the volumetric flow rate of $River_{out}$ in $\frac{m^3}{day}$.
- C_{out}: the concentration of pollution of $River_{out}$ in $\frac{kg}{m^3}$.

Exercise A

1/1 point (graded)
Five quantities have been defined: Q_{in}, C_{in}, V_{lake}, Q_{out}, and C_{out}. The values of four of these quantities influence the amount of pollutant in the lake. They have to be included in the mathematical model for $Z(t)$.

Which quantity does **not** influence Z?

- ○ Q_{in}
- ○ C_{in}
- ○ V_{lake}
- ○ Q_{out}
- ◉ C_{out} ✔

Answer
Correct:
$River_{out}$ has the same concentration pollutant as in the lake, so $C_{out} = Z/V_{lake}$. The value of C_{out} is a consequence of Z, not a cause.

Submit You have used 1 of 2 attempts

Answers are displayed within the problem

Exercise B

1/1 point (graded)
We also have to quantify how much pollutant the factory dumps into the lake. We will use just one quantity for that: U.

What is the most appropriate unit for U?

- kg
- kg per day ✔
- liter per day
- pound per day

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Answers are displayed within the problem

In the next section, you will derive a mathematical model for $Z(t)$: another differential equation.